MIL-A-13092F
24 October 1985
SUPERSEDING
MIL-A-13092E
6 August 1979

MILITARY SPECIFICATION

ADAPTER SET, ENGINE ELECTRICAL TEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adapter set for use in testing sealed automotive type electrical circuits.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571	-	Solder, Tin Alloy: Lead-Tin Alloy; and Lead Alloy.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

MIL-T-152 - Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment.
MIL-F-14256 - Flux, Soldering, Liquid (Rosin Base).

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

* 2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY

7540877 - Adapter, Primary Circuit.
7540884 - Adapter, Spark Plug.
7540895 - Adapter, Ignition Unit.
7541200 - Adapter, Testing Regulator.
7541209 - Adapter, Testing Generator.
7541281 - Adapter Set, Wheeled Vehicle.
7541361 - Data, Instruction, Adapter Set.
7541362 - Box, Adapter Set.
7723494 - Electrical Connections, Waterproof.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

* 3.2 Design. The adapter set shall conform to US Army Drawing A7541281. Adapters with pin or socket contacts shall consist of items with the contact arrangements shown on Drawing K7723494. The adapters shall be fabricated so that neither pin nor socket contacts will be damaged when mating the adapters to their plug or receptacle. The use of adapter sets furnished under this specification shall not adversely affect engine operation. When required (see 6.2.1), individual adapters, or the box, shall be furnished as specified by the procuring activity (see Table I).

TABLE I. Adapters and box.

Drawing No.	Part No.	Nomenclature
B7540877	7540877	Adapter, Primary Circuit
B7540884	7540884	Adapter, Spark Plug
B7540895	7540895	Adapter, Ignition Unit
B7541200	7541200	Adapter, Testing Regulator
C7541209	7541209	Adapter, Testing Generator
C7541362	7541362	Box, Adapter Set

* 3.2.1 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130. Components of the same part number shall be interchangeable (see 3.8.1).

* 3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3 Materials. Materials shall be as specified on the drawings. Materials not specifically designated shall be suitable for use in the testing of electrical circuits on military vehicles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

* 3.3.1 Plastic. Plastic having a cellulose nitrate base shall not be used in fabrication of the adapters. Should any plastic material used in fabrication burn after being removed from a flame that ignited it, the burning rate shall be less than two inches per minute.

3.3.2 Solder and flux. Solder used in fabrication of the adapters shall conform to Composition Sn60 of QQ-S-571. Flux used in soldering operations shall conform to MIL-F-14256. All excess flux shall be removed after soldering.

* 3.4 Construction. The adapter shells shall have a smooth clean finish to prevent damage to surfaces of mating plugs and receptacles. The bonding of mating parts on cemented adapters shall have a tensile strength that shall withstand a force of not less than 250 pounds per square inch. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. Such parts subject to adjustment or removal shall not be swaged, peened, staked, or otherwise deformed.

3.5 Performance.

3.5.1 Voltage drop (total circuit resistance). With equipment stabilized at +65° Fahrenheit (F) to +75°F, resistance of the individual circuits shall be such that the potential drop at the specified current shall not exceed the applicable values specified below:

US Army Drawing B7540884	-	20 millivolts at 5 amperes.
US Army Drawing B7540895	-	30 millivolts at 35 amperes.
US Army Drawing B7541200	-	100 millivolts at 60 amperes.
US Army Drawing C7541209	-	100 millivolts at 60 amperes (armature circuit).
US Army Drawing C7541209	-	45 millivolts at 10 amperes (field circuit).

* 3.5.2 High potential. The adapters used only in the primary circuit of ignition systems shall show no evidence of smoking, arcing, flash-over, charred insulation, or other damage when subjected to the application of not less than 500 volts root mean square (rms) alternating current (ac), 60 hertz (Hz), sine wave, for not less than one minute. In addition, adapters employed in the engine secondary circuit shall withstand not less than 10,000 volts rms ac, 60 Hz, sine wave, without sustaining the damage identified above.

3.5.3 Durability. Each adapter tested shall show no damage, and shall meet the circuit resistance requirements specified herein, after being subjected to 1000 cycles of engagement and separation.

3.5.4 Alignment of pins and sockets. Alignment of pins and sockets shall be concentric with each other within .006 inch total indicator reading. Pins and sockets shall be parallel within .006 inch per .50 inch of contact length.

* 3.6 Fungus control. Components that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Case. The case (box) shall be in accordance with Drawing 7541362 and shall be provided with each set of adapters. Clamps shall be attached to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

inside bottom of the case to secure the individual adapters. Design and location of the clamps shall be at the manufacturer's option. The cover of the carrying case shall incorporate provision for storage of the technical data required by Drawing 7541361.

3.8 Product identification.

* 3.8.1 Identification marking. The cover of the adapter case shall be marked for identification. The identification shall include the National Stock Number, nomenclature, manufacturer's name, manufacturer's model number, and contract number. Individual adapters shall be marked with their applicable part numbers as specified on the drawings.

3.8.2 Identification of contacts. Contacts and terminal connections shall be marked or coded for continuity to assure proper polarity with plugs and receptacles.

3.9 Workmanship. Standards of workmanship shall assure that the adapters shall have the durability, stability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Examination (see 4.5).
(b) Circuit resistance test (see 4.6.1).
(c) High potential test (see 4.6.2).
(d) Bonding test (see 4.6.3).
(e) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	2.0
Circuit resistance test	II	1.0
High potential test	II	1.0
Bonding test	II	1.0
Packaging inspection	S-4	4.0

NOTE: Lot size shall be no more than 100 adapters or 100 adapter sets, as applicable.

* 4.5 Product examination. Visually and manually examine each adapter or adapter set to determine conformance with the requirements of 3.2 through 3.4, 3.5.4, and 3.8 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the adapters or adapter sets have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Circuit resistance test. The adapters or adapter set, as applicable, and the test apparatus shall be stabilized in an ambient air temperature of +70° ±5°F prior to performing this test. Adapters shall then be installed in a test circuit and the terminal-to-terminal voltage drop shall be determined to verify compliance with 3.5.1.

4.6.2 High potential test. The adapters shall be connected to a suitable voltage source and tested to determine compliance with 3.5.2.

4.6.3 Bonding test. Cemented adapters shall be tested on an apparatus capable of creating a tensile force of 250 pounds per square inch to determine compliance with 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

4.6.4 Plastic flammability test. A test specimen 0.50 inch in diameter and three inches in length shall be used for this test. The specimen shall be of the same material used to manufacture the components. The specimen for test shall be clamped in a support with the plastic extended outward. Under the plastic, clamp a piece of 10 mesh Bunsen burner gauze, about five inches square, in a horizontal position 0.25 inch below the bottom of the plastic, with 0.50 inch of the plastic extending beyond the edge of the gauze. A Bunsen burner with an efficient flame 0.50 to 0.75 inch in height shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the plastic. At the end of 30 seconds the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the plastic after the removal of the Bunsen burner. In case the plastic does not continue to burn after the first ignition, the burner shall be replaced under the free end for a period of 30 seconds immediately following the extinction of the flame. If the plastic does not continue to burn until the flame has reached the full length of the plastic after the second ignition, the sample shall be considered self-extinguishing. When tested as described, the plastic shall conform to the burning rate specified in 3.3.2.

4.6.5 Durability test. The adapters and connectors shall be subjected to 1000 cycles of engagement and separation in a manner which duplicates, as closely as possible, the use of the adapters under service conditions. The same ends of the adapters shall be mated throughout the 1000 cycles. The adapters shall then be inspected for evidence of wear or damage and subjected to the circuit resistance test in 4.6.1 (see 3.5.1 and 3.5.3). Failure of any unit of the sample shall be cause for rejection of the production lot.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with any applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. Bare ferrous metal surfaces on the metal case that have not been painted or plated, or any bare component of the adapter set, shall be coated with type P-1 preservative of MIL-P-116.

5.1.1.4 Unit protection. When no preservative is required, the unit protection for individual adapters or adapter sets shall be in accordance with method III of MIL-P-116. In case a preservative is required, the unit protection shall be method I of MIL-P-116 as specified in 5.1.1.3. All components, when applicable, shall be placed in the clamp holders on the inside of the case, and cushioned to prevent damage. Cushioning material shall conform to PPP-C-843. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

cushioning shall be secured with tape conforming to PPP-T-60, type optional. The case enclosing the adapter set shall be placed in a fiberboard box conforming to PPP-B-636, class weather resistant. Closure shall be in accordance with the container specification.

5.1.1.5 Technical data. Technical data, when furnished, other than the instruction data (see Drawing 7541361) shall be packaged in accordance with method IC-1 of MIL-P-116. The data shall be positioned in a secure location on the inside of the unit container (see 5.1.1.4).

* 5.1.2 Industrial. Industrial preservation shall be in accordance with ASTM D3951-82.

* 5.2 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

5.2.1 Level A. A quantity of adapters or adapter sets, packaged as specified in 5.1.1.4, shall be packed in a shipping container conforming to PPP-B-621, class 2. Arrangement, cushioning, bracing, and blocking shall be in accordance with MIL-STD-1186.

* 5.2.2 Level B. The adapters shall be packed as specified in 5.2.1 except the containers shall conform to PPP-B-621, class 1.

* 5.2.3 Industrial. Industrial packing shall be in accordance with ASTM D3951-82.

* 5.3 Marking. Marking shall be Military or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military marking is required, it shall be so specified in the contract.

* 5.3.1 Military. Military marking shall be in accordance with MIL-STD-129.

* 5.3.2 Industrial. Industrial marking shall be in accordance with ASTM D3951-82.

6. NOTES

* 6.1 Intended use. The adapters covered by this specification are used to adapt test equipment to the sealed electrical circuits of wheeled military vehicles for trouble shooting and testing. The following electrical circuits and equipment are tested by the adapters covered by this specification:

(a) Battery to regulator circuit
(b) Spark plugs
(c) Primary circuit
(d) Generator
(e) Ignition unit

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Individual adapters or box when required (see 3.2).
(d) First article inspection if different (see 4.3).
(e) Quality conformance inspection if different (see 4.4).
(f) Level of preservation, packing, and marking if different (see 5.1, 5.2, and 5.3).

* 6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - AL
Navy - YD

Review activity:

Navy - MC

Preparing activity:

Army - AL

Project No. 4910-0601

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-13092, Rev F	ADAPTER SET, ENGINE ELECTRICAL TEST

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 02 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:03Z
Check the source to verify that this is the current version before use.